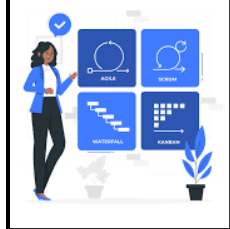


**PRESERVING ARCHITECTURAL STYLES IN THE
SEARCH-
BASED DESIGN OF SOFTWARE PRODUCT LINE
ARCHITECTURES**

Elaborado por: Ariadna Cardona, Julián Quiroz y
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2025

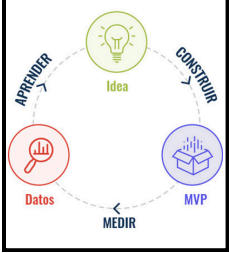
**EMBED STYLE CONSTRAINTS IN THE
SEARCH**

Architectural styles should be represented as explicit constraints in the search space, ensuring that only solutions which respect the structural rules of the selected architectural style are considered valid during optimization.



**MODEL WITH STYLE
AWARENESS**

Architectural models must include formal definitions of architectural styles, allowing the system to recognize, validate, and enforce those styles throughout the software product line design process.



**STYLE-AWARE EVOLUTIONARY
ALGORITHMS**

Evolutionary algorithms, such as genetic algorithms, can be adapted to respect architectural style constraints, generating optimized architectures that are both high-quality and stylistically consistent.



**AUTOMATED DESIGN WITH
STYLE GUIDANCE**

Search-based design automation should be guided by established architectural principles, allowing automatic generation of architectures that follow consistent and well-defined structural patterns.



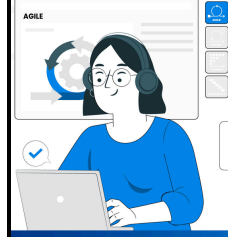
VALIDATE STYLE COMPLIANCE

Every candidate architecture must be evaluated not only on performance or cost but also on how well it adheres to the intended architectural style, using formal validation methods and style-specific metrics.



BALANCE STYLE AND TRADE-OFFS

Design decisions must carefully balance adherence to architectural styles with competing goals such as flexibility, scalability, or resource efficiency—highlighting the importance of trade-off analysis.

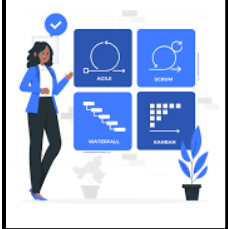


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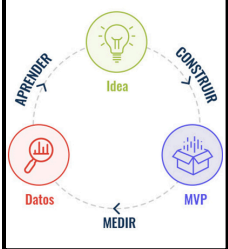
ENHANCE MODULARITY VIA STYLES

Preserving architectural styles contributes to achieving modularity, code reuse, and ease of maintenance, which are foundational for long-term sustainability in software product lines.



OPTIMIZE FOR ARCHITECTURAL QUALITY

Architectural quality attributes—such as low coupling, high cohesion, and well-defined layering—should be treated as optimization objectives, ensuring both style compliance and technical soundness.



SUPPORT CONTROLLED EVOLUTION

By enforcing consistent architectural styles, teams can evolve product lines in a controlled and predictable way, avoiding structural erosion and ensuring compatibility across versions.



TACKLE SCALABILITY AND INTEGRATION

Combining search-based techniques with architectural constraints brings challenges like performance at scale and integration with modeling tools, which must be addressed for real-world adoption.



DEFINE STYLES EARLY

Architectural styles must be defined early in the modeling and design process to guide decisions and ensure consistency across all derived products in the software product line.



AUTOMATE STYLE VALIDATION

Use automated tools and validation techniques to check architectural style compliance throughout the iterative design process, reducing manual verification effort and increasing design reliability.

